Finly: Controlled Delegation for an Alpaca Trading Agent

One autonomous strategy with a four-fund allocation sleeve, a capped-risk options sleeve, and restart-safe paper execution

Bruce Wen · Brandeis University · bwen412@brandeis.edu

2 September 2026

Abstract

Finly is one autonomous competition strategy built for the Alpaca paper-trading environment. It coordinates a diversified four-fund allocation sleeve with a capped-risk SPY options sleeve while keeping each sleeve's signals, risk account, and broker lifecycle independently auditable. The design begins from a narrow observation: interpreting information and authorizing financial risk are different jobs. A language model may be useful for reading current public news, but it should not silently acquire the authority to choose an instrument, enlarge a position, alter a loss bound, or manufacture a broker order. Finly therefore separates model judgment from capital-bearing decisions and records the boundary in machine-checkable artifacts.

The four-fund sleeve is internally identified as G4. It assigns one half of its research allocation to QQQ and divides the other half among the three strongest original Select Sector SPDR funds under a twelve-to-six-month relative-momentum rule. The competition deployment scales that allocation to 97 percent, preserves a 3 percent cash reserve, and performs one initial rebalance without in-contest re-optimization. In a consumed, post-selected simulation from 2 January 2013 through 27 August 2026, with modeled one-way costs of five basis points, G4 returned 967.11 percent while SPY returned 580.82 percent. The result is economically large but did not pass Finly's promotion gate: its Deflated Sharpe probability was 3.75 percent, its worst familywise-adjusted bootstrap p-value was 37.18 percent, it did not consistently beat a static SPY/QQQ growth control, and the frozen Yahoo-versus-Alpaca source reconciliation failed closed.

The coordinated SPY options sleeve uses the frozen long-only SPY/BIL receipt as a freshness and long-horizon risk-on guard, not as the short-horizon trade direction. In live policy v2, fresh SPY price momentum owns the bullish or bearish sign. Option skew and Qwen3-32B's assessment of canonical public Alpaca news may confirm, reduce, or veto that signed view but cannot reverse it. Code enumerates defined-risk bull-call or bear-put debit spreads, evaluates two 2,048-path scenario models, applies conservative expected-value and probability gates, limits live entry to one contract and at most $500 of loss, and issues a thirty-second HMAC-signed v3 risk certificate bound to one exact order projection. Alpaca execution proceeds through a fresh preflight, durable state machine, deterministic client-order identifiers, pre-mutation checkpoints, broker read-back, and fail-closed reconciliation.

This paper specifies the algorithms, causal timing, cost conventions, statistical tests, risk constraints, state transitions, and reproducibility boundary. Through the September 2 closing bell, 15 ETF fill events had built the allocation sleeve; Finly was up $141.24 while the same-$100,000 SPY raw-price benchmark was down $284.76, a $426.00 advantage at one exact timestamp. The options sleeve completed 24 live evaluation cycles that day; all ended NO_TRADE, so it opened no option position and added no new options risk. The evidence supports an inspectable paper-trading experiment, not durable alpha, verified live options profit, or future outperformance.

1. Problem, scope, and contributions

Trading-agent demonstrations often collapse a chain of unlike decisions into one prompt. A model reads a report, states a view, chooses a product, chooses a size, and emits something shaped like an order. That sequence is attractive on screen because it is short. It is also difficult to audit. A mistaken interpretation, a stale quote, a sizing error, and a duplicated broker mutation have different causes and require different controls.

Finly treats the agent as a controlled pipeline rather than a single oracle. The project makes four contributions.

1. **One strategy with coordinated sleeves.** G4 supplies a frozen four-fund allocation for the time-bounded competition account. The capped-risk options sleeve uses a distinct signed live policy instead of inferring direction from the G4 backtest; separate signals and risk accounts preserve auditability without creating a second competition strategy.
2. **Signed price authority with reduction-only confirmation.** Fresh market momentum owns the short-horizon sign. Option skew and the hosted model's bounded news assessments may confirm, reduce, or veto that view, but cannot reverse it, choose an option, choose quantity, access account secrets, or authorize a broker mutation.
3. **A deterministic compiler and permit.** Typed evidence becomes a checked intent; an enumerator constructs vertical spreads; two scenario models evaluate them; fixed gates select at most one candidate; and an HMAC-signed certificate binds evidence, policy, risk, account, market, candidate, quantity, and order projection.
4. **Restart-safe Alpaca execution.** The cloud runner persists intent before mutation, advances at most one broker-changing step per cycle, assigns deterministic client-order identifiers, reconciles ambiguous acknowledgements, and publishes only sanitized status.

Research, synthetic demonstration, paper authorization, and broker observation are distinct. A replay is not a fill, a fixture is not an executable permit, and a paper fill is not real-capital execution. Tests establish software behavior, not forecast accuracy [1–7].

1.1 Versioned live-policy provenance

The G4 allocation rule, historical ledger, frozen SPY/BIL economic research, and first-session measurements were not revised. After the original options policy produced repeated NO_TRADE decisions, Finly activated bidirectional live policy v2 at **3:28:21 UTC** and a prospective asymmetric-payoff calibration at **4:08:18 UTC on 2 September 2026**. The cloud workflow in revision `d8e1fcc` pins implementation `318c943`: fresh market momentum may select a bounded bullish or bearish view, while the frozen economic receipt remains a freshness and risk-on guard. A candidate must show positive conservative value, at least 1.25-to-1 reward/risk, and at least 45 percent modeled probability of profit. The $500 loss ceiling, 0.5-percent equity budget, one-contract limit, paper-only endpoint, HMAC permit, final preflight, idempotent order identity, and broker reconciliation remained hard controls. Every earlier decision and P&L observation remains attributed to the policy revision that produced it. No live options fill is claimed.

2. Architecture and authority

Finly coordinates two execution paths inside one competition strategy. They meet at the same paper account and measurement layer while retaining distinct state and risk records.

```
Historical adjusted closes
    │
    ├── G4 research rule ── frozen 2026-08-28 signal
    │                 │
    │                 └── one-time QQQ/XLB/XLE/XLV plan
    │                          │
    │                     signed equity state machine
    │                          │
    │                     Alpaca MCP stock orders
    │
Alpaca SPY/BIL bars ── frozen freshness + risk-on guard ──────────┐
Alpaca SPY market ───── price-momentum sign: bull / bear / neutral ├─┐
Alpaca option chain ── deterministic confirmation / reduction ──┤  │
Alpaca public news ──── Qwen confirmation / reduction / veto ───┘  │
                                    ▼
                          checked economic intent
                                    │
                       enumerate and value SPY vertical spreads
                                    │
                    source-removal + 32-case alpha diagnostics
                                    │
                        HMAC risk certificate + fresh preflight
                                    │
                             Alpaca MCP multi-leg paper order

Both sleeves ── broker read-back ── encrypted state ── sanitized feed/scorer
```

Component	May decide	May not decide
G4 research rule	Ranked sectors and research weights	Options, credentials, or discretionary overrides
Frozen competition protocol	Dated one-time allocation and 3% reserve	In-contest re-optimization
Frozen SPY/BIL economic receipt	Freshness and long-horizon risk-on gate	Short-horizon sign, contract, quantity, or broker payload
Market-momentum policy	Bullish, bearish, or neutral short-horizon sign	Contract, quantity, credentials, or mutation
Options and Qwen confirmation	Bounded confirmation, reduction, or veto	Sign reversal, contract, size, credentials, mutation
Deterministic compiler	Eligible spread and order-independent risk	Broker submission
Certificate and preflight	Permit or no trade for one projection	General permission to trade
Alpaca executor	Submit or reconcile the bound paper order	Change certified economics

The threat model assumes public text can be misleading, duplicated, stale, or adversarial; reads can fail; acknowledgements can be lost after acceptance; scheduled jobs can begin late; and state can be modified. Historical prices, paper fills, and a short competition window are not treated as proof of long-run alpha.

3. The frozen G4 equity sleeve

3.1 Research rule

G4 operates on QQQ, BIL as the simulator's cash symbol, and the original nine Select Sector SPDR funds: XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, and XLY. On each eligible twenty-one-session signal date it computes twelve-to-six-month momentum:

$$m_s(t) = \ln[P_s(t-126) / P_s(t-252)].$$

The six-month gap excludes the most recent 126 sessions from the ranking interval. The strategy sorts sectors by descending $m_s(t)$, breaks an exact tie alphabetically, chooses the first three, assigns 50 percent to QQQ, and assigns one sixth to each selected sector. All remaining weights are zero. It is long-only with risky gross exposure capped at one [8].

```
Algorithm 1: G4 research allocation

Inputs:
 adjusted closes P for QQQ, BIL, and nine sector funds
 lookback L = 252 sessions
 gap endpoint G = 126 sessions
 rebalance interval R = 21 sessions
 anchor a in {0, ..., 20}

Initialize the portfolio in BIL.

For each eligible signal close t:
 If (t − L − a) mod R = 0:
  For each sector s:
   momentum[s] ⟵ ln(P[s,t−G] / P[s,t−L])
  selected ⟵ top three by descending momentum,
       then ascending symbol for exact ties
  target[QQQ] ⟵ 0.50
  target[s] ⟵ 1/6 for each selected sector
  target[all other risky symbols] ⟵ 0
  target[BIL] ⟵ 0
 Else:
  retain the drifted portfolio

 Queue a new target at close t.
 Execute it at close t+1.
 First earn its return from close t+1 to close t+2.
```

The momentum literature motivates delayed trend measurement but does not validate this universe, lookback, gap, or core weight [9]. G4 was selected after substantial research on the available ETF history and is post-selected.

3.2 Competition transform

The competition signal is distinct from a recurring G4 backtest. Alpaca IEX adjustment-all daily bars through 28 August 2026 supplied 253 retained sessions and selected XLB, XLE, and XLV [2]. The protocol scaled research weights by 0.97:

$$w_{QQQ} = 0.97 \times 0.50 = 0.485; \quad w_{XLB} = w_{XLE} = w_{XLV} = 0.97 \times (1/6) = 0.1616667; \quad w_{cash} = 0.03.$$

For the exact \$100,000 baseline, target notionals are \$48,500 for QQQ and approximately \$16,166.67 for each sector. The protocol fixes the signal session, source, adjustment, targets, paper endpoint, MCP version, order method, identifier namespace, and artifact hashes. It authorizes one initial rebalance and no in-contest re-optimization [3]. A human chose this dated research candidate for the competition; the live account is its forward observation, not a continuation of the backtest.

4. Causal simulation and benchmark methodology

4.1 Data and timing

The headline G4 replay uses a hash-pinned panel of Yahoo Finance adjusted closes. The runner retained aligned sessions and did not impute unavailable 28 August 2026 adjusted-close rows, so the uniform panel ends on 27 August [4,5]. The competition signal was reproduced separately from Alpaca IEX bars. A provider-overlap experiment later failed its strict reconciliation gate; the two sources must not be described as equivalent [6].

A decision formed with observations through close t is queued at t, executes at close t+1, and first earns the return from t+1 to t+2. Two later closes must exist before a signal is eligible. This prevents the close used to form a rank from also earning the return into that close.

Let $w^*_i(t)$ be a queued target, $w^-_i(t+1)$ the drifted holding immediately before execution, and $R_i(t+1,t+2)$ the subsequent return. One-way turnover and cost are

$$\tau(t+1) = \Sigma_i |w^*_i(t) - w^-_i(t+1)|; \quad C(t+1) = \tau(t+1) \times b / 10{,}000.$$

Gross and net returns are

$$g(t+2) = \Sigma_i w_i(t+1)R_i(t+1,t+2); \quad r(t+2) = g(t+2) - C(t+1) - \max[0, -w\_BIL(t+1)]s\_borrow/252.$$

G4 is unlevered, so financing cost is zero. Holdings then drift:

$$w^-_i(t+2) = w_i(t+1)[1 + R_i(t+1,t+2)] / [1 + g(t+2)].$$

Every standalone date slice is rebased from cash. Its first row is charged entry from BIL and its last row is charged terminal liquidation. SPY buy-and-hold receives the identical panel, lag, entry, and terminal-cost treatment [5].

4.2 Metrics and partitions

For n sessions and initial wealth V_0,

$$V\_n = V_0 \Pi_t[1 + r(t)]; \quad \text{Annualized return} = (V\_n/V_0)\hat{}(252/n) - 1; \quad \text{Annualized volatility} = \sqrt{252} \times \text{sample standard deviation}[r(t)].$$

$$\text{Cash-excess Sharpe} = \sqrt{252} \times \text{mean}[r(t) - r\_BIL(t)] / \text{sample standard deviation}[r(t) - r\_BIL(t)];$$
$$\text{Maximum drawdown} = \min_t \{V(t)/\max_{u\le t}[V(u)] - 1\}.$$

Five-percent daily expected shortfall averages observations at or below the empirical fifth percentile. These ledger metrics are not probabilities of future profit.

The robustness protocol reports development from 2 June 2008 through 29 December 2017, validation from 2 January 2018 through 31 December 2024, and a consumed recent diagnostic from 2 January 2025 through 28 August 2026. Validation was observed during selection and the recent interval was consumed by a safety veto. No pristine holdout remained [6].

Comparators included SPY, QQQ, the lower-volatility frozen Finly policy, a 15-percent-volatility SPY target, and a static 50/50 SPY/QQQ control. That static control is necessary because half of G4 is always QQQ; sector rotation that cannot consistently beat a simple growth tilt cannot establish independent sector-selection alpha.

5. Historical evidence and statistical audit

5.1 ETF-era result

The release gate permits one precise headline. In the consumed, post-selected 2 January 2013 through 27 August 2026 replay, with five basis points of modeled one-way cost, G4 returned 967.11 percent and SPY returned 580.82 percent [4].

Modeled $10,000 investment	Total return	Ending wealth
G4	+967.11%	$106,711
SPY, identical panel and cost convention	+580.82%	$68,082

Modeled $10,000 investment	Total return	Ending wealth
Historical ending-wealth difference	—	$38,629

The public wealth and drawdown series permit chart reproduction [7]. These values accurately report the frozen ledger under its assumptions. They are not an unbiased estimate of expected excess return.

5.2 Robustness that passed

The raw SPY edge stayed positive at five, ten, and twenty-five basis points. The schedule experiment tested every possible offset of the twenty-one-session cycle. These are sensitivity variants, not independent samples.

Annualized net-log-growth edge over SPY	Minimum	Median	Maximum	Positive offsets
Development, 2008–2017	1.17 pp	1.79 pp	3.32 pp	21/21
Validation, 2018–2024	2.28 pp	2.96 pp	4.40 pp	21/21

Thus the result is not explained by one convenient rebalance weekday. This is useful falsification evidence, but it does not correct strategy search or establish independence from growth exposure.

5.3 Deflated Sharpe and familywise tests

For candidate j and SPY on the same validation row,

$$\Delta_j(t) = r_j(t) - r\_SPY(t); \quad T_j = \sqrt{n} \times \text{mean}[\Delta_j(t)].$$

The Deflated Sharpe calculation uses paired-return skewness, Pearson kurtosis, and a declared cumulative trial count of 100. G4's annualized paired Sharpe was 0.8614 while the trial-adjusted benchmark was 1.5330. The estimated probability that the observed Sharpe exceeded the deflated benchmark was 0.03748, below the frozen 0.95 gate [6,10].

The White-style maximum-statistic bootstrap centers candidate-minus-SPY returns under the null. Each iteration applies one shared index path to all seven candidate series. For iteration b,

$$T^{max}(b) = \max_j \{\sqrt{n} \times \text{mean}[\Delta^0{}_j,b(t)]\}; \quad p\_FWER = [1 + \text{count}(T^{max}(b) \geq T\_G4)] / (B + 1).$$

The protocol freezes B = 2,000 iterations for circular and moving blocks of 5, 20, and 60 sessions. Unadjusted fixed-candidate p-values were approximately 0.0045 to 0.0090, but familywise p-values were roughly 0.30 to 0.37. The worst was 0.371814; all six failed the five-percent gate.

Promotion gate	Threshold	Observed	Result
Deflated Sharpe probability	≥ 95%	3.75%	Fail
All six familywise p-values	≤ 5%	Worst 37.18%	Fail
SPY edge at 5/10/25 bp	Positive at all costs	Positive	Pass
Edge at all 21 offsets	Positive in development and validation	21/21, 21/21	Pass
Consistency above static 50/50 SPY/QQQ	Frozen rolling-window rule	Not consistent	Fail
Authenticated source overlap	Every used symbol passes	Fail-closed	Fail

The disposition is “retrospective raw-return evidence; not statistically promoted.” It motivates a dated experiment, not future-profit language [33,34].

5.4 Earlier industry proxy

A separate fixed replay uses Kenneth French's value-weighted ten-industry daily portfolios. The primary window spans 7 May 1927 through 29 May 2007, 21,218 observations, outside the modern ETF-era overlap. These are industry portfolios, not tradable QQQ and SPDR shares, so the exercise tests the shape of an industry-momentum proxy [11,12].

1927–2007 modeled comparison	Finly proxy	Market proxy
Annualized return	13.37%	9.48%
Cash-excess Sharpe	0.573	0.429
Maximum drawdown	−67.76%	−84.07%
Annualized net-log-growth advantage	3.49 pp	—

All twenty-one offsets retained a positive edge. The edge remained 2.45 percentage points at a twenty-five-basis-point cost, and maximum drawdown was 16.31 percentage points shallower. A volatility-matched market comparison still left a 2.17-percentage-point net-log-growth advantage.

Five of seven complete decades were positive. A nominal one-sided bootstrap p-value of 0.0024 became 0.4824 after Bonferroni adjustment for 201 global trials:

$$p\_adjusted = \min(1,\ 201 \times 0.0024) = 0.4824.$$

The Deflated Sharpe probability was 0.7182. Eight of nine precommitted gates passed, but the overall statistical gate failed. This supplies cross-era economic evidence, not independent proof of durable alpha.

6. Frozen economic research and the live risk-on guard

G4 chooses the competition equity sleeve; it does not authorize options. A separate frozen long-only SPY/BIL policy called tsmom_ensemble_vol supplies a reproducible long-horizon economic receipt [13,14]. Live v2 reuses that checked receipt only as a freshness and risk-on guard; it does not use the receipt to choose bullish versus bearish short-horizon direction [36].

For h in {21, 63, 252}, it computes SPY's excess log trend:

$$x\_h(t) = \ln[SPY(t)/SPY(t-h)] - \ln[BIL(t)/BIL(t-h)].$$

Let q(t) be the fraction of positive trends and $\sigma_{20}(t)$ the annualized sample volatility of twenty SPY daily returns:

$$q(t) = (1/3)\ \Sigma\_h\ I[x\_h(t) > 0];\quad v(t) = \min[1,\ 0.10/\sigma_{20}(t)];\quad w\_SPY(t) = q(t)v(t);\quad w\_BIL(t) = 1 - w\_SPY(t).$$

The frozen research policy is unlevered, never shorts, and proposes a rebalance every five completed sessions. A daily bar becomes eligible only after regular close plus fifteen minutes, and SPY/BIL sessions must align. In research, a close-t signal executes at t+1 and first earns t+1-to-t+2 returns. In deployment it produces a hashed economic receipt and does not mutate the broker.

In the separate 2 January 2025 through 28 August 2026 execution-realism artifact, it returned 15.39 percent at five basis points per traded leg and 10.56 percent at twenty-five basis points, versus SPY's 33.52 percent. At five basis points it had 8.12 percent modeled annualized volatility and −5.45 percent maximum drawdown. It was risk-controlled, not market-beating on total return [4].

For an options entry, the guard requires a fresh checked receipt and at least 50 percent indicated SPY exposure. If it fails, the outcome is no trade. If it passes, the current live path derives bounded directional capacity from the checked SPY weight,

$$u(t) = 0.5 + 0.5|w\_SPY(t)-0.5|,$$

but leaves the short-horizon sign to fresh SPY price momentum. This preserves the frozen research artifact without converting its long-only allocation into a bullish-only options rule.

7. Evidence, model role, and deterministic intent

7.1 Canonical signals

Every evidence record includes family, underlying, source kind and URI, origin ID, publication and receipt timestamps, content hash, duplicate-group hash, and an evidence ID derived from the canonical body. IDs, origins, and duplicate groups cannot cross families. A second signal from one family is rejected rather than double-counted [15,16].

The deployed runner builds three live families.

- **Market.** With twenty recent SPY log returns and volatility σ:

$$z\_market = 0.65\ M_5/(\sigma\sqrt{5}) + 0.35\ M_{20}/(\sigma\sqrt{20}); \quad d\_market = \tanh(z\_market/2).$$

- **Options.** At the earliest expiry with both rights, code selects nearest-to-spot call and put. For κ = IV_put − IV_call:

$$d\_options = -\tanh(\kappa/0.035).$$

- **Events.** Up to twelve nonduplicated Alpaca news items no older than seventy-two hours receive age weights exp(−age_hours/24); the weighted model score is shrunk by 0.65.

The schema also supports a prediction-market family with base weight 0.05, and the complete synthetic fixture exercises it. The current autonomous live builder has no production prediction-market feed. It is a supported family, not a current live source.

7.2 Hosted model boundary

Qwen3-32B receives only canonical public news text, required evidence IDs, underlying, and timestamp. Temperature is zero, extended thinking is disabled, and output must contain exactly one assessment per ID: direction score, volatility score, and short rationale. Unknown keys, missing or duplicate IDs, nonfinite scores, malformed JSON, timeout, identity mismatch, or hash mismatch fails extraction [17].

The model has no account identifier, credential, equity, buying power, position, compiler state, risk object, candidate list, order payload, or mutation tool. Source text is untrusted data, not instructions. Model output is evidence assessment, not a trade.

7.3 Signed market authority and reduction-only confirmation

For family j,

$$a\_j = b\_j \times quality\_j \times freshness\_j \times calibration\_j \times independence\_j,$$

where base weights are 0.50 market, 0.20 options, 0.25 events, and 0.05 prediction market. In the generic library path,

$$d = \Sigma_j a_j d_j / \Sigma_j a_j.$$

Coverage is $\min(1, \Sigma a_j)$; agreement is the share of absolute directional weight aligned with the permitted sign. The generic frozen library path retains thresholds $|d| \geq 0.18$, coverage ≥ 0.35, and agreement ≥ 0.55. Live v2 uses $|d| \geq 0.12$, coverage ≥ 0.30, and agreement ≥ 0.45 [16,18].

In the current live path, market momentum alone owns the sign. Let $s = \mathrm{sign}(d_{market})$, and let u be the capacity from the frozen risk-on guard. Options and event evidence that oppose s create

$$g_{confirm} = \mathrm{clip}[1 - \Sigma_{j\in\{options,events\}, \mathrm{sign}(d_j)=-s} \{|d_j|a_j/b_j\}, 0, 1].$$

The supported score is

$$d_{supported} = s \times \min[u, |d_{market}|] \times g_{confirm}.$$

Thus a bullish market score can compile only a bull-call spread, a bearish market score can compile only a bear-put spread, and neutral or insufficiently confirmed evidence becomes no trade. Options skew and Qwen-assessed news may improve agreement or reduce the view; neither can reverse the price sign or enlarge its magnitude above the market-and-guard envelope. This signed, reduction-only relationship is Finly's central controlled-delegation property.

8. Deterministic options compilation

8.1 Enumeration and payoff

The compiler accepts SPY and three library outcomes: bull-call debit spread, bear-put debit spread, or no trade. Quotes must pass schema, underlying, right, feed, age, relative spread, open-interest, tradability, and DTE checks. Entry DTE is three to fourteen days; maximum relative leg spread is 0.25; minimum open interest is 50; indicative quotes may be ninety seconds old and OPRA quotes thirty seconds [18].

For each same-expiry pair, code assigns the lower-strike call as bullish long leg or higher-strike put as bearish long leg. Width W is \$1–\$15. Entry debit is

$$D = ask_{long} - bid_{short} + 2(\$0.03).$$

Candidates require $0 < D < W$. For one contract,

$$\text{Maximum loss} = 100D; \quad \text{Maximum gain} = 100(W - D).$$

with live-policy reward-to-risk at least 1.25. Terminal payoff is

$$\Pi_{call}(S_T) = 100\{\min[W, \max(0, S_T - K_{long})] - D\}; \quad \Pi_{put}(S_T) = 100\{\min[W, \max(0, K_{long} - S_T)] - D\}.$$

8.2 Scenario valuation

Every spread is evaluated under two models with 2,048 paths [19,20]. The tilted-implied model uses normal quantiles. For horizon h, direction d, volatility score v, and mean leg IV:

$$\sigma_A = \max[0.08, IV(1 + 0.12v)]; \quad \ln[S_h/S_0] = (0.38d - \sigma_A^2/2)(h/252) + \sigma_A\sqrt{(h/252)}z.$$

The second model draws five-session circular blocks from historical SPY log returns. Its target daily volatility is

$$\sigma\_B{,}daily = [0.55\sigma\_hist + 0.45(IV/\sqrt{252})](1 + 0.12v).$$

Centered historical returns are scaled to that value, restored to their mean, and receive directional tilt 0.00115d per session. Seeds derive deterministically from candidate ID and market time.

Remaining option time at the horizon is

$$T\_remaining = \max[1/365, DTE/365 - h/252].$$

Each leg is marked with Black–Scholes [21]:

$$d_1 = \{\ln(S/K) + (r - q + \sigma^2/2)T\} / (\sigma\sqrt{T});\quad d_2 = d_1 - \sigma\sqrt{T};\quad C = Se\hat{}(-qT)\Phi(d_1) - Ke\hat{}(-rT)\Phi(d_2);\quad P = Ke\hat{}(-rT)\Phi(-d_2) - Se\hat{}(-qT)\Phi(-d_1).$$

Close value is clipped to [0,W] after two $0.03 exit-slippage charges. For model m, Finly records mean profit μ_m, standard error SE_m, probability of profit, and five-percent expected shortfall. Conservative expected value is

$$CEV = \min\_m\ [\mu\_m - 1.645\ SE\_m];\quad \text{the live policy requires } CEV \geq \max[\$5, 0.02 \times \text{maximum loss}], \text{ conservative probability} \geq 0.45, \text{ and reward/risk} \geq 1.25.$$

Probability and expected shortfall take the worse model. Among candidates with loss no greater than $500, the compiler maximizes CEV divided by maximum loss and breaks exact ties by stable candidate hash.

8.3 Size and challenge suite

With equity E and per-contract loss L, the live policy permits

$$Q = 1 \text{ if } L \leq \min(0.005E, \$500), \text{ and } Q = 0 \text{ otherwise.}$$

There is no live half-risk or multi-contract branch. New plus open defined risk may not exceed 0.03E [22]. Thus every live entry is one contract, fits a 0.5-percent equity budget, and has no more than $500 of certified maximum loss.

The complete synthetic fixture contains four families, so leave-one-family-out testing creates four variants. In the current live policy these variants are alpha diagnostics: every nonneutral variant must retain the base sign, and an opposite-sign flip is prohibited. Neutralization is recorded as lost confidence rather than a hard-safety failure.

Thirty-two deterministic Halton variants perturb source direction by ±0.05, quality by ±0.03, freshness and calibration by ±0.02, independence by ±0.015, spot by ±0.2 percent, leg IV by ±4 percent, history scale from 0.94 to 1.06, interest rate by ±0.5 percentage points, horizon by −1/0/+1, and entry debit by less than $0.04 [23]. Live diagnostics require zero opposite-sign flips, at least 75 percent nonneutral direction, 50 percent trade rate, 50 percent same structure, and nonnegative nearest-rank fifth-percentile CEV. These statistics describe alpha confidence. They do not authorize broker mutation.

In the frozen published synthetic run, four of four removals passed; all thirty-two variants retained direction and trade; trade and same-structure rates were 100 percent; and fifth-percentile CEV was $71.48. The selected bearish fixture had $3.66 debit, $366 maximum loss, $634 maximum gain, CEV $97.18, and 60.89 percent conservative probability [24]. It remains a synthetic compiler test under the generic replay policy. Its synthetic_replay certificate cannot authorize the paper executor, and its figures are not evidence of a live options fill.

9. Certificate and paper execution

9.1 One permit for one order

A paper-submit risk_certificate.v3 is valid for thirty seconds. It contains run and expiry times, paper mode, scope, intent hash, candidate ID and snapshot hash, desired order-projection hash, policy hash, evidence root, account and market hashes, observed spot and time, feed, quantity, loss reservation, entry ceiling, account equity, open risk, CEV, probability, expected shortfall, diagnostic summaries, and all Boolean checks [22]. The hard certificate checks require an eligible candidate, exactly one contract, maximum loss within $500 and the 0.5-percent budget, aggregate risk within three percent of equity, a fresh recognized quote feed, a fresh unblocked paper account, and MCP execution transport. Source-removal and perturbation results are recorded as alpha diagnostics rather than hard authorization checks.

The certificate ID hashes its canonical body. A separate secret of at least thirty-two bytes signs it:

```
signature = HMAC-SHA256(secret, canonical certificate body).
```

Verification recomputes body hash and HMAC with timing-safe equality, checks paper mode and paper_submit scope, and rejects stale permits. Synthetic certificates use a separate scope. Final preflight may be fifteen seconds old at most; reviewed debit may drift $0.10 and SPY 0.5 percent; the account needs Level 3 options and at least 25 percent post-trade buying power. Preflight, idempotent client-order identity, pre-mutation state persistence, and broker reconciliation remain mandatory after certificate issuance.

9.2 Broker state machine

The equity runner persists signed phases PLANNED, ORDER_PENDING, RECONCILING, READY, and FROZEN [25].

```
Algorithm 2: one conservative G4 cloud cycle

Restore and authenticate encrypted state.
Reject a mismatched protocol, revision, account binding, or signature.

If READY:
  verify exact holdings, quantities, and no open equity order
  return READY or freeze a hard contradiction
If FROZEN:
  return without mutation

Read clock, account, positions, orders, and asset eligibility.
Require the dedicated active, unblocked, initially flat $100,000 paper account,
the exact market window, paper endpoint, and mutation acknowledgements.

Choose the first leg not FILLED.
If no leg remains:
  reconcile total notional, cash reserve, exact symbols, and quantities
  defer while broker endpoints converge
  otherwise transition RECONCILING → READY

Before placing one leg:
  persist ORDER_PENDING with mutation_started = true
  compare-and-swap the state revision and checkpoint encryption
  call Alpaca MCP once

If acknowledgement is absent:
  look up the deterministic client-order ID
  if still absent, return ORDER_AMBIGUOUS without resubmission

Read back the order.
Pending stays ORDER_PENDING.
Failure or contradiction becomes FROZEN.
Fill signs quantity, average price, and notional,
then returns to PLANNED or advances to RECONCILING.
```

Each cycle permits at most one broker-changing call. Deterministic IDs hash the frozen protocol, sequence, symbol, and notional. Because intent is durable before mutation, a runner dying after Alpaca accepts an order recovers the same ID rather than creating a duplicate.

Final readiness requires filled notional no lower than $20 below the authorized total and no higher than 97 percent of baseline plus one cent; cash no lower than 3 percent minus $5; exactly QQQ, XLB, XLE, and XLV; no open equity order; and broker quantities matching signed fills. Temporary endpoint convergence defers; arithmetic or identity contradiction freezes.

The options lifecycle is separate. At the force-flat deadline, Finly reconciles the exact working close, cancels only that order, waits for terminal cancellation, and permits at most one deterministic replacement at a $0.01 credit floor. Lost cancellation acknowledgements and restarts recover by exact client-order identifier. Finly never expands the close into a debit or market order; the floor bounds behavior but cannot guarantee a fill.

9.3 Cloud state and measurement

GitHub Actions runs while the laptop is off. Schedule time is not authority: an in-process UTC window and Alpaca clock govern mutation. Concurrency permits one cycle. The official Alpaca MCP server is pinned at 2.2.1 [26,27].

Lifecycle journals are encrypted with AES-256-GCM under a secret different from the certificate secret and stored on an isolated state branch. State is checkpointed before entry and exit mutation. The public branch contains only allowlisted competition_live.json without credentials, raw account identity, or broker IDs. GitHub Pages fetches that file; it never receives a trading secret [27].

The GET-only scorer compares Finly with same-timestamp SPY and rejects cashflow or provenance breaks [28]. Through the September 2 close, Finly was +$141.24 versus SPY -$284.76 at 4:00 p.m. ET: +$426.00 after 15 ETF fill events and zero external cashflows [35]. The same day's 24 live options evaluations produced 14 certification failures, six insufficient-evidence decisions, and four post-cutoff decisions; all 24 ended NO_TRADE with zero option orders and fills [38].

9.4 Current eligibility calibration

The separately versioned live options policy has a deterministic reachability check [37]. It samples 517 five-session SPY signal windows from 20 May 2016 through 26 August 2026, each using 96 completed daily returns, and compiles them against one fixed, symmetric 14-DTE, 20-percent-IV modeled quote surface at a common spot. Eleven of 517 windows (2.12766 percent) clear every gate: seven bullish call spreads and four bearish put spreads. Their maximum losses are $440–$455, conservative EVs are $10.08–$22.26 against required margins of $8.80–$9.10, reward/risk ratios are 2.2967–2.4091, and modeled probabilities of profit are 0.4512–0.4771. The artifact binds the 2,679 Alpaca adjusted closes and all policy inputs by hash.

This calibration establishes that the path is reachable but selective. It is not an options P&L backtest: the option surfaces are fixed modeled sensitivities, not historical quotes, and no broker order, fill, or return is inferred.

10. Verification, reproducibility, and hackathon fit

The public run executed 827 tests: 825 passed, none failed, and two were skipped [29]. Coverage includes schemas, evidence separation, G4 weights, causal lag, costs, signed live direction, eligibility-calibration integrity, option payoff, Black–Scholes, scenario determinism, removal and perturbation diagnostics, live one-contract risk limits, v3 HMAC scope, broker translation, stale preflight, bounded cancel/reprice exits, duplicate recovery, encryption, calendar boundaries, forward attribution, and artifact checks. This establishes tested software behavior, not forecast accuracy or profitability.

The repository pins Node.js 26.7.0. A clean reproduction begins:

```
npm ci
npm run verify
```

The command runs linting, tests, replay and receipt checks, recorded model-decision checks, economic research, current-decision and economic-options checks, registered forward-trial verification, MCP configuration, production build, and submission validation. Historical evidence, gates, receipts, protocol, forward contract, and sanitized status are machine-readable JSON. References point to a fixed revision.

Finly maps to the Options Alpha Agents requirements [1,30].

Requirement	Implementation	Boundary
Autonomous AI agent using Alpaca	Hosted assessment, economic intent, compiler, certificate, lifecycle, cloud runner	Interpretation is not broker authority
Alpaca MCP or CLI	Official MCP 2.2.1, paper endpoint, authenticated path	Read access is not a fill
Incorporate options	SPY vertical compiler, payoff, limits, multi-leg translation	ETF history is not options P&L
Dedicated $100,000 paper account	Active Level 3 paper account with private ID and sanitized snapshot [32]	Credentials stay outside Git
One-page explanation	Separate judge brief plus this technical paper	The paper supplements the brief
Public original work	MIT repository and source-bound artifacts [31]	Secrets and private state excluded

The strongest claim is not that an LLM found a guaranteed trade. It is that the division of authority is executable and versioned: market momentum chooses a bounded sign; options and model evidence may confirm or reduce it; code computes the exact spread and risk; the v3 permit binds one projection; Alpaca records allocation execution; and a public scorer measures the account without retroactively rewriting earlier results.

11. Threats and limitations

Selection and multiplicity. G4 followed extensive exploration. ETF-era partitions were consumed, statistical promotion failed, and no pristine holdout remains. The external proxy also failed its adjusted gate. Historical outperformance is not a forecast.

Data mismatch. G4 research uses Yahoo adjusted closes; competition signaling uses Alpaca IEX adjustment-all. Strict reconciliation failed for every required symbol. Adjusted series reflect vendor methodology and revised corporate actions.

Universe and control. The ETF universe consists of 2026 survivors. QQQ creates a growth tilt, and rotation did not consistently beat static SPY/QQQ. French industries are proxies, not the modern instruments.

Execution. Close replays with fixed costs omit queue priority, dynamic spreads, partial fills, halts, latency, impact, and some corporate-action cases. Alpaca paper fills cannot establish real-capital execution quality.

Options model risk. Positive CEV inside two approximate models can fail under jumps, regime change, surface movement, or unmodeled dependence. Defined risk does not make direction correct.

News model. Qwen can misunderstand or omit context. Schema validation controls shape, not truth; confirmation-only authority limits the consequence but does not establish forecast accuracy.

Post-diagnostic calibration. Live v2 was calibrated after the original configuration repeatedly abstained. Its direction and confidence thresholds therefore have little independent forward exposure. The revision is disclosed and versioned, but it is not evidence of proven options alpha.

Operations. GitHub Actions can fail or run late. State, timing, read, or reconciliation failures become no trade or frozen state.

Short forward window. A few days cannot estimate long-run return. One realized path primarily tests operational execution.

12. Conclusion

Finly is one autonomous competition strategy with two coordinated execution sleeves; it does not pretend that their models, signals, or risk accounts are interchangeable. The four-fund sleeve supplies a compelling historical reason to test: $10,000 became $106,711 versus $68,082 for SPY. The failed statistical, control, and source gates supply the reason to be careful.

The system turns that tension into design. The frozen SPY/BIL receipt supplies a freshness and long-horizon risk-on guard without choosing the live sign. Fresh SPY price momentum owns that sign; option skew and news may confirm, reduce, or veto but cannot reverse it. Code owns payoff, EV gates, one-contract quantity, loss, and broker fields. A short-lived v3 HMAC certificate binds one state to one projection, while final preflight, encrypted checkpoints, deterministic IDs, and broker reconciliation make uncertain acknowledgements recoverable. The paper account then observes each versioned rule without retroactively changing earlier results.

Finly's contribution is controlled delegation: every component has a useful job, every capital-bearing decision has a deterministic owner, and every public claim traces to supporting evidence.

References

[1] Finly. “Submission Requirements and Release Checklist.” docs/REQUIREMENTS_MATRIX.md.
[2] Finly. “Frozen G4 Source Signal.” config/g4-official-source-signal.json.
[3] Finly. “Frozen G4 Competition Protocol.” config/g4-official-production.json.
[4] Finly. “Quantitative Release Gate.” research/output/quantitative_release_gate.json.
[5] Finly. “Generation 4 Simulation Engine.” research/champion_engine.mjs.
[6] Finly. “Generation 4 Robustness Result.” research/output/quant_champion_generation4_robustness.json.
[7] Finly. “G4 Wealth and Drawdown Series.” public/data/g4_wealth_drawdown.json.
[8] Finly. “Generation 4 Strategy Definitions.” research/champion_strategies_generation4.mjs.

[9] T. J. Moskowitz, Y. H. Ooi, and L. H. Pedersen. “Time Series Momentum.” *Journal of Financial Economics* 104, no. 2 (2012): 228–250. doi:10.1016/j.jfineco.2011.11.003.

[10] Finly. “Generation 4 Statistical Methods.” research/champion_statistics.mjs.

[11] Kenneth R. French. Data Library.

[12] Finly. “External-Era Public Evidence.” public/data/attempt150_public_evidence.json.

[13] Finly. “Frozen Economic Research Policy.” lib/economic_research.mjs.

[14] Finly. “Economic Research Evidence.” public/data/economic_research.json.

[15] Finly. “Live Signal Construction.” lib/live_signals.mjs.

[16] Finly. “Signed Evidence Aggregation.” lib/signals.mjs.

[17] Finly. “Hosted Evidence Extractor.” lib/evidence_extractor.mjs.

[18] Finly. “Options, Live Alpha, and Safety Policy.” lib/policy.mjs.

[19] Finly. “Deterministic Options Compiler.” lib/compiler.mjs.

[20] Finly. “Scenario and Black–Scholes Functions.” lib/quant.mjs.

[21] F. Black and M. Scholes. “The Pricing of Options and Corporate Liabilities.” *Journal of Political Economy* 81, no. 3 (1973): 637–654. doi:10.1086/260062.

[22] Finly. “v3 Risk Certificate.” lib/risk.mjs.

[23] Finly. “Source-Removal and Perturbation Diagnostics.” lib/stability.mjs.

[24] Finly. “Latest Synthetic Options Decision Receipt.” public/data/latest_receipt.json.

[25] Finly. “Official G4 Equity Runner.” lib/g4_official_equity.mjs.

[26] Alpaca. “MCP Server.” Official documentation.

[27] Finly. “Cloud Runner and Restart Design.” docs/CLOUD_RUNNER.md.

[28] Finly. “Competition Forward-Profit Contract.” config/competition-forward-profit.json.

[29] Finly. “Public Automated Verification.” GitHub Actions.

[30] lablab.ai. Alpaca AI Trading Agents Hackathon.

[31] Finly. Public repository.

[32] Finly. “Sanitized Competition Account Snapshot.” public/data/competition_live.json.

[33] H. White. “A Reality Check for Data Snooping.” *Econometrica* 68, no. 5 (2000): 1097–1126. doi:10.1111/1468-0262.00152.

[34] D. H. Bailey and M. López de Prado. “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality.” *Journal of Portfolio Management* 40, no. 5 (2014): 94–107. doi:10.3905/jpm.2014.40.5.094.

[35] Finly. “September 2 Same-Clock Forward Measurement.” public/data/competition_forward_profit_2026_09_02.json.

[36] Finly. “Live v2 Economic Options Authority.” lib/live_economic_options_authority.mjs.

[37] Finly. “Current Signal and Quote-Surface Eligibility Calibration.” public/data/options_policy_calibration.json.

[38] Finly. “September 2 Live Options Decision Record.” public/data/options_live_decision_funnel_2026_09_02.json.